Parameter Name	Description	Parameter Value	Description
FEEDBACK_PATH	Selects the feedback path to the PLL	SIMPLE	Feedback is internal to the PLL, directly from VCO
		DELAY	Feedback is internal to the PLL, through the Fine Delay Adjust Block
		PHASE_AND_DELAY	Feedback is internal to the PLL, through the Phase Shifter and the Fine Delay Adjust Block
		EXTERNAL	Feedback path is external to the PLL, and connects to EXTFEEDBACK pin. Also uses the Fine Delay Adjust Block.
DELAY_ADJUSTMENT_MODE_FEEDBACK	Selects the mode for the Fine Delay Adjust block in the feedback path	FIXED	Delay of the Fine Delay Adjust Block is fixed, the value is specified by the FDA_FEEDBACK parameter setting
		DYNAMIC	Delay of Fine Delay Adjust Block is determined by the signal value at the DYNAMICDELAY[3:0] pins
FDA_FEEDBACK	Sets a constant value for the Fine Delay Adjust Block in the feedback path	0, 1,…,15	The PLLOUTGLOBALA & PLLOUTCOREA signals are delay compensated by (n+1)*150 ps, where n = FDA_FEEDBACK only if the setting of the DELAY_ADJUSTMENT_MODE_FEEDBACK is FIXED.
DELAY_ADJUSTMENT_MODE_RELATIVE	Selects the mode for the Fine Delay Adjust block	FIXED	Delay of the Fine Delay Adjust Block is fixed, the value is specified by the FDA_RELATIVE parameter setting
		DYNAMIC	Delay of Fine Delay Adjust Block is determined by the signal value at the DYNAMICDELAY[7:4] pins
FDA_RELATIVE	Sets a constant value for the Fine Delay Adjust Block	0, 1,…,15	The PLLOUTGLOBALA & PLLOUTCOREA signals are delayed w.r.t. the Port B signals, by (n+1)*150 ps, where n = FDA_RELATIVE. Used if DELAY_ADJUSTMENT_MODE_RELATIVE is "FIXED".
SHIFTREG_DIV_MODE	Selects shift register configuration	0,1	Used when FEEDBACK_PATH is "PHASE_AND_DELAY". 0→Divide by 4 1→Divide by 7
PLLOUT_SELECT_PORTA	Selects the signal to be output at the PLLOUTCOREA and PLLOUTGLOBALA ports	SHIFTREG_0deg	0^{o} phase shift only if the setting of FEEDBACK_PATH is "PHASE_AND_DELAY"
		SHIFTREG_90deg	90^{o} phase shift only if the setting of FEEDBACK_PATH is "PHASE_AND_DELAY" and SHIFTREG_DIV_MODE=0
		GENCLK	The internally generated PLL frequency will be output to PortA. No phase shift.
		GENCLK_HALF	The internally generated PLL frequency will be divided by 2 and then output to PORTA. No phase shift.

PLLOUT_SELECT_PORTB	Selects the signal to be output at the PLLOUTCOREB and PLLOUTGLOBALB ports	SHIFTREG_0deg	0^o phase shift only if the setting of FEEDBACK_PATH is “PHASE_AND_DELAY”
		SHIFTREG_90deg	90^o phase shift only if the setting of FEEDBACK_PATH is “PHASE_AND_DELAY” and SHIFTREG_DIV_MODE=0
		GENCLK	The internally generated PLL frequency will be output to PortB. No phase shift.
		GENCLK_HALF	The internally generated PLL frequency will be divided by 2 and then output to PORTB. No phase shift.
DIVR	REFERENCECLK divider	0,1,2,…,15	These parameters are used to control the output frequency, depending on the FEEDBACK_PATH setting.
DIVF	Feedback divider	0,1,..,63	
DIVQ	VCO Divider	1,2,…,6	
FILTER_RANGE	PLL Filter Range	0,1,…,7	
EXTERNAL_DIVIDE_FACTOR	Divide-by factor of a divider in external feedback path	User specified value. Default 1	Specified only when there is a user-implemented divider in the external feedback path.
ENABLE_ICEGATE_PORTA	Enables the PLL power-down control	0	Power-down control disabled
		1	Power-down controlled by LATCHINPUTVALUE input
ENABLE_ICEGATE_PORTB	Enables the PLL power-down control	0	Power-down control disabled
		1	Power-down controlled by LATCHINPUTVALUE input

Hard Macro Primitives

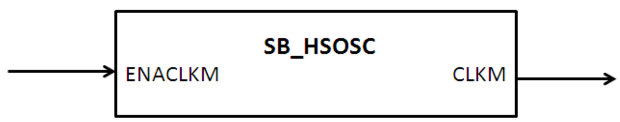
iCE40LM Hard Macros

This section describes the following dedicated hard macro primitives available in iCE40LM devices.

SB_HSOSC (macro primitive for HSSG)
SB_LSOSC (macro primitive for LPSG)
SB_I2C
SB_SPI

SB_HSOSC (For HSSG)

SB_HSOSC primitive can be used to instantiate High Speed Strobe Generator (HSSG), which generates 12 MHz strobe signal. The strobe can drive either the global clock network or fabric routes directly based on the clock network selection.



Ports

SB_HSOSC Ports		
Signal Name	**Direction**	**Description**
ENACLKM	Input	Enable High Speed Strobe Generator. Active High.
CLKM	Output	Strobe Generator Output (12Mhz).

Clock Network Selection

By default the strobe generator use one of the dedicated clock networks in the device to drive the elements. The user may configure the strobe generator to use the fabric routes instead of global clock network using the synthesis attributes.

Synthesis Attribute

/* synthesis ROUTE_THROUGH_FABRIC=<value> */

Value:

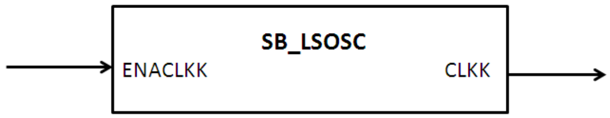
0: Use dedicated clock network. Default option.
1: Use fabric routes.

Verilog Instantiation

```
SB_HSOSC OSCInst0 (
      .ENACLKM(ENACLKM),
      .CLKM(CLKM)
      ) /* synthesis ROUTE_THROUGH_FABRIC= [0|1] */;
```

SB_LSOSC (For LPSG)

SB_LSOSC primitive can instantiate Low Power Strobe Generator (LPSG), which generates 10 KHz strobe signal. The strobe can drive either the global clock network or fabric routes directly based on the clock network selection.



Ports

SB_LSOSC Ports		
Signal Name	**Direction**	**Description**
ENACLKK	Input	Enable Low Power Strobe Generator. Active High.
CLKK	Output	Strobe Generator Output (10Khz).

Clock Network Selection

By default the strobe generator use one of the dedicated clock networks in the device to drive the elements. The user may configure the strobe generator to use the fabric routes instead of global clock network using the synthesis attribute.

Synthesis Attribute:

/* synthesis ROUTE_THROUGH_FABRIC=<value> */

Value:

0: Use dedicated clock network. Default option.
1: Use fabric routes.

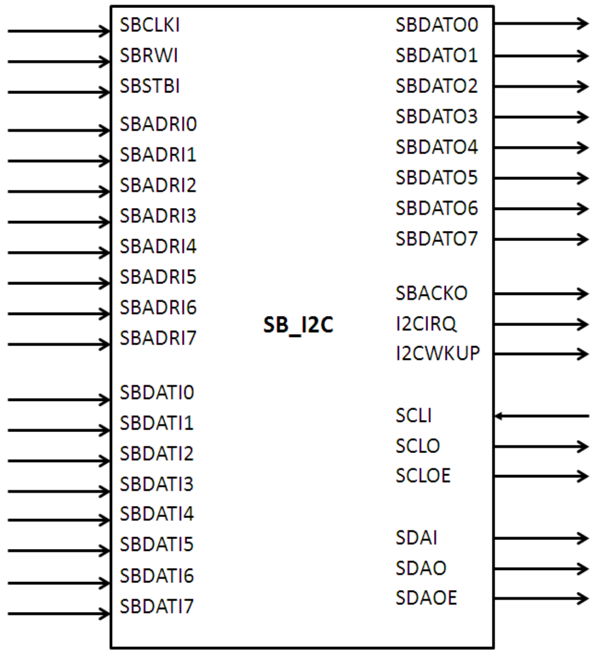
Verilog Instantiation

```
SB_LSOSC OSCInst0 (
      .ENACLKK(ENACLKK),
      .CLKK(CLKK)
      ) /* synthesis ROUTE_THROUGH_FABRIC= [0|1] */;
```

SB_I2C

The I2C hard IP provides industry standard two pin communication interface that conforms to V2.1 of the I2C bus specification. It could be configured as either master or slave port. In master mode, it support configurable data transfer rate and perform arbitration detection to allow it to operate in multi-master systems. It supports both 7 bits and 10 bits addressing in slave mode with configurable slave address and clock stretching in both master and slave mode with enable/disable capability.

iCE40LM device supports two I2C hard IP primitives , located at upper left corner and upper right corner of the chip.



Ports

SB_I2C Ports

Signal Name	Direction	Description
SBCLKI	Input	System Clock input.
SBRWI	Input	System Read/Write Input.
SBSTBI	Input	Strobe Signal
SBADRI0	Input	System Bus Control registers address. Bit 0.
SBADRI1	Input	System Bus Control registers address. Bit 1.
SBADRI2	Input	System Bus Control registers address. Bit 2.
SBADRI3	Input	System Bus Control registers address. Bit 3.
SBADRI4	Input	System Bus Control registers address. Bit 4.
SBADRI5	Input	System Bus Control registers address. Bit 5.
SBADRI6	Input	System Bus Control registers address. Bit 6.
SBADRI7	Input	System Bus Control registers address. Bit 7.
SBDATI0	Input	System Data Input. Bit 0.
SBDATI1	Input	System Data input. Bit 1.
SBDATI2	Input	System Data input. Bit 2.
SBDATI3	Input	System Data input. Bit 3.
SBDATI4	Input	System Data input. Bit 4.
SBDATI5	Input	System Data input. Bit 5.
SBDATI6	Input	System Data input. Bit 6.
SBDATI7	Input	System Data input. Bit 7.
SBDATO0	Output	System Data Output. Bit 0.
SBDATO1	Output	System Data Output. Bit 1.
SBDATO2	Output	System Data Output. Bit 2.
SBDATO3	Output	System Data Output. Bit 3.
SBDATO4	Output	System Data Output. Bit 4.
SBDATO5	Output	System Data Output. Bit 5.
SBDATO6	Output	System Data Output. Bit 6.
SBDATO7	Output	System Data Output. Bit 7.

SBACKO	Output	System Acknowledgement.
I2CIRQ	Output	I2C Interrupt output.
I2CWKUP	Output	I2C Wake Up from Standby signal.
SCLI	Input	Serial Clock Input.
SCLO	Output	Serial Clock Output
SCLOE	Output	Serial Clock Output Enable. Active High.
SDAI	Input	Serial Data Input
SDAO	Output	Serial Data Output
SDAOE	Output	Serial Data Output Enable. Active High.

Parameters

I2C Primitive requires configuring certain parameters for slave initial address and selecting I2C IP location.

I2C Location	Parameters	Parameter Default Value.	Description.
Upper Left Corner	I2C_SLAVE_INIT_ADDR	0b1111100001	Upper Bits <9:2> can be changed through control registers. Lower bits <1:0> are fixed.
	BUS_ADDR74	0b0001	Fixed value. SBADRI [7:4] bits also should match with this value to activate the IP.
Upper Right Corner	I2C_SLAVE_INIT_ADDR	0b1111100010	Upper Bits <9:2> can be changed through control registers. Lower bits <1:0> are fixed.
	BUS_ADDR74	0b0011	Fixed value. SBADRI [7:4] bits also should match with this value to activate the IP.

Synthesis Attribute

Synthesis attribute “I2C_CLK_DIVIDER” is used by PNR and STA tools for optimization and deriving the appropriate clock frequency at SCLO output with respect to the SBCLKI input clock frequency.

/* synthesis I2C_CLK_DIVIDER=[Divide Range] */

Divide Range : 0, 1, 2, 3 … 1023. Default is 0.

Verilog Instantiation

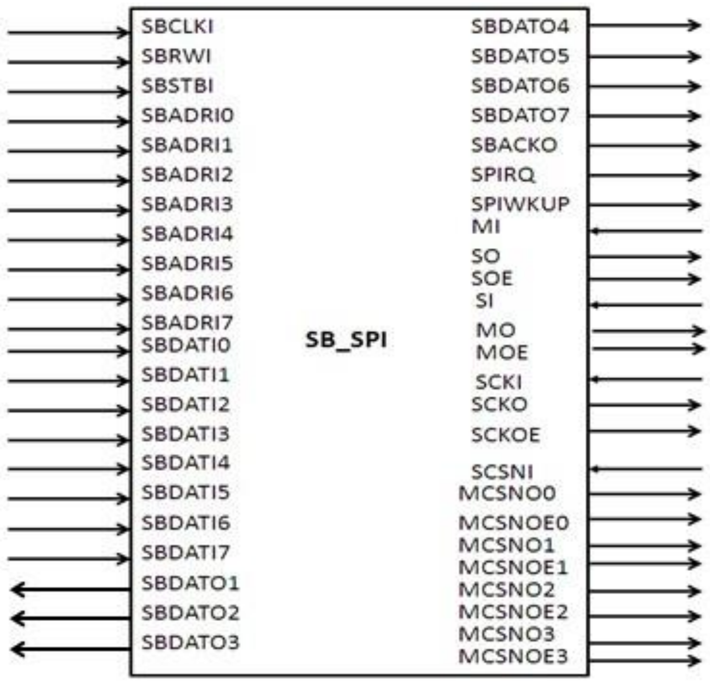
```
SB_I2C i2cInst0 (
    .SBCLKI(sbclki),
    .SBRWI(sbrwi),
    .SBSTBI(sbstbi),
    .SBADRI7(sbadri[7]),
    .SBADRI6(sbadri[6]),
    .SBADRI5(sbadri[5]),
    .SBADRI4(sbadri[4]),
    .SBADRI3(sbadri[3]),
    .SBADRI2(sbadri[2]),
    .SBADRI1(sbadri[1]),
    .SBADRI0(sbadri[0]),
    .SBDATI7(sbdati[7]),
    .SBDATI6(sbdati[6]),
    .SBDATI5(sbdati[5]),
    .SBDATI4(sbdati[4]),
    .SBDATI3(sbdati[3]),
    .SBDATI2(sbdati[2]),
    .SBDATI1(sbdati[1]),
    .SBDATI0(sbdati[0]),
    .SCLI(scli),
    .SDAI(sdai),
    .SBDATO7(sbdato[7]),
    .SBDATO6(sbdato[6]),
    .SBDATO5(sbdato[5]),
    .SBDATO4(sbdato[4]),
    .SBDATO3(sbdato[3]),
    .SBDATO2(sbdato[2]),
    .SBDATO1(sbdato[1]),
    .SBDATO0(sbdato[0]),
    .SBACKO(sbacko),
    .I2CIRQ(i2cirq),
    .I2CWKUP(i2cwkup),
    .SCLO(sclo),
    .SCLOE(scloe),
    .SDAO(sdao),
    .SDAOE(sdaoe)
 )/* synthesis I2C_CLK_DIVIDER= 1 */;

defparam i2cInst0.I2C_SLAVE_INIT_ADDR = "0b1111100001";
defparam i2cInst0.BUS_ADDR74 = "0b0001";
```

SB_SPI

The SPI hard IP provide industry standard four-pin communication interface with 8 bit wide System Bus to communicate with System Host. It could be configured as Master or Slave SPI port with separate Chip Select Pin. In master mode, it provides programmable baud rate, and supports CS HOLD capability for multiple transfers. It provides variety status flags, such as Mode Fault Error flag, Transmit/Receive status flag etc. for easy communicate with system host.

iCE40LM device supports two SPI hard IP primitives, located at lower left corner and lower right corner of the chip.



Ports

SB_SPI Ports		
Signal Name	**Direction**	**Description**
SBCLKI	Input	System Clock input.
SBRWI	Input	System Read/Write Input.
SBSTBI	Input	Strobe Signal
SBADRI0	Input	System Bus Control registers address. Bit 0.
SBADRI1	Input	System Bus Control registers address. Bit 1.
SBADRI2	Input	System Bus Control registers address. Bit 2.
SBADRI3	Input	System Bus Control registers address. Bit 3.
SBADRI4	Input	System Bus Control registers address. Bit 4.
SBADRI5	Input	System Bus Control registers address. Bit 5.
SBADRI6	Input	System Bus Control registers address. Bit 6.
SBADRI7	Input	System Bus Control registers address. Bit 7.
SBDATI0	Input	System Data Input. Bit 0.
SBDATI1	Input	System Data input. Bit 1.
SBDATI2	Input	System Data input. Bit 2.
SBDATI3	Input	System Data input. Bit 3.
SBDATI4	Input	System Data input. Bit 4.
SBDATI5	Input	System Data input. Bit 5.
SBDATI6	Input	System Data input. Bit 6.
SBDATI7	Input	System Data input. Bit 7.
SBDATO0	Input	System Data Output. Bit 0.
SBDATO1	Input	System Data Output. Bit 1.
SBDATO2	Input	System Data Output. Bit 2.
SBDATO3	Input	System Data Output. Bit 3.
SBDATO4	Input	System Data Output. Bit 4.
SBDATO5	Input	System Data Output. Bit 5.
SBDATO6	Input	System Data Output. Bit 6.

SBDATO7	Input	System Data Output. Bit 7.
SBACKO	Output	System Acknowledgement
SPIIRQ	Output	SPI Interrupt output.
SPIWKUP	Output	SPI Wake Up from Standby signal.
MI	Input	Master Input from PAD
SO	Output	Slave Output to PAD
SOE	Output	Slave Output Enable to PAD. Active High.
SI	Input	Slave Input from PAD
MO	Output	Master Output to PAD
MOE	Output	Master Output Enable to PAD. Active High
SCKI	Input	Slave Clock Input From PAD
SCKO	Output	Slave Clock Output to PAD
SCKOE	Output	Slave Clock Output Enable to PAD. Active High.
SCSNI	Input	Slave Chip Select Input From PAD
MCSNO0	Output	Master Chip Select Output to PAD. Line 0.
MCSNO1	Output	Master Chip Select Output to PAD. Line 1.
MCSNO2	Output	Master Chip Select Output to PAD. Line 2.
MCSNO3	Output	Master Chip Select Output to PAD. Line 3.
MCSNOE0	Output	Master Chip Select Output Enable to PAD. Active High. Line 0.
MCSNOE1	Output	Master Chip Select Output Enable to PAD. Active High. Line 1
MCSNOE2	Output	Master Chip Select Output Enable to PAD. Active High. Line 2
MCSNOE3	Output	Master Chip Select Output Enable to PAD. Active High. Line 3

Parameters

SPI Primitive requires configuring a parameter for selecting the SPI IP location.

I2C Location	Parameters	Parameter Default Value.	Description.
Lower Left Corner	BUS_ADDR74	0b0000	Fixed value. SBADRI [7:4] bits also should match with this value to activate the IP.
Lower Right Corner	BUS_ADDR74	0b0001	Fixed value. SBADRI [7:4] bits also should match with this value to activate the IP.

Synthesis Attribute

Synthesis attribute “SPI_CLK_DIVIDER” is used by PNR and STA tools for optimization and deriving the appropriate clock frequency at SCKO output with respect to the SBCLKI input clock frequency.

/* synthesis SPI_CLK_DIVIDER= [Divide Range] */

Divide Range : 0, 1, 2, 3….63. Default is 0.

Verilog Instantiation

```
SB_SPI spiInst0 (
    .SBCLKI(sbclki),
    .SBRWI(sbrwi),
    .SBSTBI(sbstbi),
    .SBADRI7(sbadri[7]),
    .SBADRI6(sbadri[6]),
    .SBADRI5(sbadri[5]),
    .SBADRI4(sbadri[4]),
    .SBADRI3(sbadri[3]),
    .SBADRI2(sbadri[2]),
    .SBADRI1(sbadri[1]),
    .SBADRI0(sbadri[0]),
    .SBDATI7(sbdati[7]),
    .SBDATI6(sbdati[6]),
    .SBDATI5(sbdati[5]),
    .SBDATI4(sbdati[4]),
    .SBDATI3(sbdati[3]),
    .SBDATI2(sbdati[2]),
    .SBDATI1(sbdati[1]),
    .SBDATI0(sbdati[0]),
    .MI(mi),
    .SI(si),
    .SCKI(scki),
    .SCSNI(scsni),
    .SBDATO7(sbdato[7]),
    .SBDATO6(sbdato[6]),
    .SBDATO5(sbdato[5]),
    .SBDATO4(sbdato[4]),
    .SBDATO3(sbdato[3]),
    .SBDATO2(sbdato[2]),
    .SBDATO1(sbdato[1]),
    .SBDATO0(sbdato[0]),
    .SBACKO(sbacko),
    .SPIIRQ(spiirq),
    .SPIWKUP(spiwkup),
    .SO(so),
    .SOE(soe),
    .MO(mo),
    .MOE(moe),
    .SCKO(scko),
    .SCKOE(sckoe),
    .MCSNO3(mcsno_hi[3]),
    .MCSNO2(mcsno_hi[2]),
    .MCSNO1(mcsno_lo[1]),
    .MCSNO0(mcsno_lo[0]),
    .MCSNOE3(mcsnoe_hi[3]),
    .MCSNOE2(mcsnoe_hi[2]),
    .MCSNOE1(mcsnoe_lo[1]),
    .MCSNOE0(mcsnoe_lo[0])
 ) /* synthesis SPI_CLK_DIVIDER = "1" */;

defparam spiInst0.BUS_ADDR74 = "0b0000";
```